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COUNT RELATIONS AND REGULAR LANGUAGES

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§0. Introduction.

The language a*b* is regular, but $\{a^n b^n \mid n \in N\}$ is not. The language $\{a^n a^n \mid n \in N\}$ is regular, but $\{(a)^n c(a)^n \mid n \in N\}$ is not. What relations between sequences of strings $x,y \in \Sigma^+$ in words $\alpha(x)^n\beta(y)^m\gamma$, $\alpha,\beta,\gamma \in \Sigma^*$, can be enforced in a regular language? In this report we provide a general answer to this question. The non-regularity of $\{a^n b^n \mid n \in N\}$, $\{(a)^n c(a)^n \mid n \in N\}$, $\{a^n b^m \mid m > n\}$, $\{a^n b^n \mid n \in N\} \cup L$, where L is any finite set, and many other languages follows immediately from our main result (Theorem 2).

Our notation is stardard: Σ is any finite alphabet; languages L are subsets of Σ^*. Λ is the null string. N is the set of all natural numbers. $|N| = \omega$ denotes the cardinality of N. Parentheses are usually included in expressions like "$(a)^n c(a)^n$" but not in expressions like "$a^n b^n$", where the exponents are clearly unambiguous.

§1. Pumps and count relations.

Following Ehrenfeucht et al. (1981) we define a pump as follows.† Let $L \subseteq \Sigma^*$, $w \in \Sigma^*$, $z \neq \Lambda$, and $w = \alpha z \beta$. Then z is a pump for w relative to L iff, for all $i \geq 0$,

$$\alpha(z)^i\beta \in L \text{ iff } w \in L.$$

As Ehrenfeucht et al. point out, being a pump is a "local" property in the sense that it does not depend on words $\alpha'(z)^i\beta'$ where $\alpha' \neq \alpha$ or $\beta' \neq \beta$. Furthermore, being a pump is a property of a particular occurrence of a string z

† Apart from notational differences, this definition is identical to the one provided by Ehrenfeucht et al. (1981); we depart from them only in requiring that $z \neq \Lambda$.

in a particular word w.

We extend and generalize this definition to a local property of occurrences of pairs of substrings. Let $L \subseteq \Sigma^*$, $w \in L$, $x \neq \Lambda$, $y \neq \Lambda$ and $w = \alpha x \beta y \gamma$. We define the count relation $R \subseteq N \times N$ for $\langle x,y \rangle$ in the word $\alpha x \beta y \gamma$ relative to L by the condition:

$$R\ (i,j) \text{ iff } \alpha(x)^i \beta(y)^j \gamma \in L.$$

Obviously, if either x or y is a pump for $\alpha x \beta y \gamma$ in L, then R is an infinite subset of $N \times N$.

It is clear that having a particular count relation R in a word $\alpha x \beta y \gamma$ is, like the property of being a pump, a property of particular occurrences of x and y in the word: occurrences of x and y in other words may have different count relations. Notice that the count relation of $\langle x,y \rangle$ in w is also relative to a given factorization of w; different occurrences of x and y in the same word may have different count relations. Thus when we say that $\langle x,y \rangle$ has count relation R in $\alpha x \beta y \gamma$ relative to L, there is an implicit relativization of the occurrences of x and y to the factorization of w given by $(\alpha,x,\beta,y,\gamma)$.

Examples. The count relation R for $\langle a,b \rangle$ in ab relative to the language $\{a^n b^n \mid n \in N\}$ is infinite; and if $a^i b^j = a^k b^l$, then $i = j = k = l$. Contrast the infinite count relation R for $\langle a,a \rangle$ in aa relative to the language a^*. Here if $a^i a^j = a^k a^l$ we know only that $i + j = k + 1$. In this case, $R = N \times N$.

Given a pair $\langle x,y \rangle$ in $\alpha x \beta y \gamma$ with count relation R (relative to L) define the sublanguage L_{xy} of L by letting

$$L_{xy} = \{w \mid w = \alpha(x)^i \beta(y)^j \gamma, \langle i,j \rangle \in R\}.$$

We shall be investigating the relation of L to such sublanguages; we first establish some properties of R.

§2. Finite decomposability.

In this section we assume we are given an arbitrary, but fixed, language L_{xy}, defined as above in terms of pair $\langle x,y\rangle$ in $\alpha x\beta y\gamma \in L$ with count relation R.

For each $i \in \mathrm{dom}R$ let

$$\Gamma_i = \{j \mid \langle i,j\rangle \in R\}$$

Notice that $i \mapsto \Gamma_i$ is a function from $\mathrm{dom}R$ into subsets of the range, $\mathrm{rng}R$, of R. Thus the relation

$$i_1 \equiv i_2 \text{ iff } \Gamma_{i_1} = \Gamma_{i_2}$$

is an equivalence relation on $\mathrm{dom}R$.

For each $i \in \mathrm{dom}R$ we define a relation

$$R_i = [i] \times \Gamma_i \qquad (*)$$

where $[i]$ is the equivalence class of i under $\equiv$.

Clearly

$$R = \cup R_i.$$

We shall say that R is <u>finitely decomposable</u> (f.d.) iff R is the union of finitely many such R_i.

<u>Lemma 1.</u> R is f.d. iff $\{\Gamma_i \mid i \in \mathrm{dom}R\}$ is finite.

<u>Proof.</u> The cardinality of $\{[i] \mid i \in \mathrm{dom}R\}$ is bounded by the cardinality of the set of distinct Γ_i. In fact they are equal, by the definition of $\equiv$. □

<u>Lemma 2.</u> The decomposition of R defined by $(*)$ is unique.

<u>Proof.</u> It suffices to show that if $i_1, i_2 \in [i]$ then $R_{i_1} = R_{i_2}$. But if $i_1, i_2 \in [i]$, $\Gamma_{i_1} = \Gamma_{i_2}$. So we need only show that $[i_1] = [i_2]$. But this follows from the fact that $\equiv$ is an equivalence relation. □

§3. Good finals and regularity.

We shall be making extensive use of Nerode's Theorem, which we state below. First we define the Nerode equivalence relation $\sim_L$ for a language L: given $x,y \in \Sigma^*$, $x \sim_L y$ iff

for all $z \in \Sigma^*$, $xz \in L$ iff $yz \in L$.

An equivalence relation is said to be of finite index or finite rank just in case the set of equivalence classes under the relation has finite cardinality.

Nerode's Theorem. A language L is regular iff $\sim_L$ is of finite index. (Nerode, 1958; See Hopcroft and Ullman, 1979, Thm. 3.9.)

For each $\sigma \in \Sigma^*$, we define the set G_σ of good finals for σ relative to L by letting

$$G_\sigma = \{z \in \Sigma^* \mid \sigma z \in L\}$$

Similarly, the set B_σ of bad finals for σ (relative to L) is given by

$$B_\sigma = \{z \in \Sigma^* \mid \sigma z \notin L\}.$$

Lemma 3. $\Sigma^*/\sim_L \approx \{G_\sigma \mid \sigma \in \Sigma^*\}$.

Proof. Clearly if $w \sim_L z$, $G_w = G_z$.

If $w \not\sim_L z$ then either $G_w \neq G_z$ or $B_w \neq B_z$. Now suppose $w \not\sim_L z$ while $G_w = G_z$. Then since $B_w = \Sigma^* - G_w$ and $B_z = \Sigma^* - G_z$, $B_w = B_z$, and so $w \sim_L z$ -- a contradiction. Thus $w \sim_L z$ iff $G_w = G_z$. Hence the map $[\sigma] \mapsto G_\sigma$ from $\Sigma^*/\sim_L$ into $\{G_\sigma \mid \sigma \in \Sigma^*\}$ is one-one, and so

$$|\Sigma^*/\sim_L| \leq |\{G_\sigma \mid \sigma \in \Sigma^*\}|.$$

Next suppose $G_w = G_z$. Then $B_w = B_z$ and so $w \sim_L z$. Similarly if $G_w \neq G_z$, then $w \not\sim_L z$. So the map $G_\sigma \mapsto [\sigma]$ from $\{G_\sigma \mid \sigma \in \Sigma^*\}$ into $\Sigma^*/\sim_L$ is

one-one. It follows that

$$|\{G_\sigma | \sigma \in \Sigma^*\}| \leq |\Sigma^*/\sim_L|,$$

and so, by Schröder-Bernstein, we have

$$\Sigma^*/\sim_L \approx \{G_\sigma | \sigma \in \Sigma^*\}. \quad \square$$

Lemma 4. $\sim_L$ is of finite index iff $|\{G_\sigma | \sigma \in \Sigma^*\}| < \omega$.

Proof. Recall that $\sim_L$ is of finite index iff there are only finitely many equivalence classes $[\sigma]$. The lemma is thus an immediate consequence of Lemma 3. $\square$

§4. Regularity and finite decomposability.

Theorem 1. Let $L_{xy} = \{\alpha(x)^i\beta(y)^j\gamma \mid \langle i,j\rangle \in R\}$ be as in §1, above. If $\sim_{L_{xy}}$ is of finite index, R is finitely decomposable.

Proof. By Lemma 4 it suffices to prove

$$|\{G_\sigma | \sigma \in \Sigma^*\}| < \omega \Rightarrow R \text{ is f.d.}$$

We shall say that δ is a maximal x-initial segment of $w = \alpha(x)^i\beta(y)^j\gamma$ iff $\delta = \alpha(x)^i\beta$ where α,β,γ are the substrings specified in the definition of L_{xy}. Let

$$\Delta = \{\delta \mid \text{for some } i \in \text{dom}R,\ \delta = \alpha(x)^i\beta\}.$$

Clearly if $\delta \in \Delta$, $\delta = \alpha(x)^i\beta$, then δ is an initial segment of a word $w \in L_{xy}$.

For each $\delta \in \Delta$, let $\exp(\delta)$ denote the exponent of x in δ. (Formally $\exp: \Delta \to N$ is the function defined by $\exp(\delta) = i$ iff $\delta = \alpha(x)^i\beta$, where α,x,β are the substrings specified in the definition of L_{xy}.) Let

$$\underset{\sim}{G} = \{G_\delta \mid \delta \in \Delta\}.$$

Since $\Delta \subset \Sigma^*$,

$$|\{G_\sigma \mid \sigma \in \Sigma^*\}| < \omega \Rightarrow |\underset{\sim}{G}| < \omega. \qquad (*)$$

Let $\underset{\sim}{\Gamma} = \{\Gamma_i \mid i = \exp(\delta),\ \delta \in \Delta\}$.

Then it suffices to show,

$$|\underset{\sim}{G}| < \omega \Rightarrow |\underset{\sim}{\Gamma}| < \omega, \qquad (**)$$

since, by Lemma 4, if $\sim_{L_{xy}}$ is of finite index, $|\{G_{\sigma} \mid \sigma \in \Sigma^*\}| < \omega$; hence by (*), $|\underset{\sim}{G}| < \omega$. Thus by (**), $|\underset{\sim}{\Gamma}| < \omega$, and so by Lemma 1, R is f.d.

We establish (**) by proving the following:

Lemma 5. $\underset{\sim}{G} \approx \underset{\sim}{\Gamma}$

Proof. Define a function h: $\underset{\sim}{G} \to \underset{\sim}{\Gamma}$ by letting $h(G_{\delta}) = \Gamma_{\exp(\delta)}$.

If $G_{\delta i} = G_{\delta j}$, where $\delta_i = \alpha x^i \beta$ and $\delta_j = \alpha x^j \beta$, then obviously $\Gamma_i = \Gamma_j$; and since for every $\Gamma_i \in \underset{\sim}{\Gamma}$, $i=\exp(\delta)$ for some $\delta \in \Delta$, h is onto. (Notice that this is sufficient to establish (**).)

To show that h is one-one, suppose $G_{\delta i} \neq G_{\delta j}$ but $\Gamma_i = \Gamma_j$. Then there is a z such that (i) $\alpha x^i \beta z \in L$ and $\alpha x^j \beta z \notin L$, or (ii) $\alpha x^i \beta z \notin L$ and $\alpha x^j \beta z \in L$. Suppose case (i) holds, and that $z = y^1 \gamma$. Then $\alpha x^i \beta y^1 \gamma \in L$ and $\alpha x^j \beta y^1 z \notin L$. Now by hypothesis $\Gamma_i = \Gamma_j$. Hence by definition of Γ, $R(i,1)$ iff $R(j,1)$, and by the definition of L_{xy},

$$R(i,1) \text{ iff } \alpha x^i \beta y^1 \gamma \in L,$$

and

$$R(j,1) \text{ iff } \alpha x^j \beta y^1 \gamma \in L.$$

Thus $\alpha x^i \beta y^1 \gamma \in L$ iff $\alpha x^j \beta y^1 \gamma \in L$-- in contradiction to (i). A similar argument applies if (ii) is assumed. Thus $\Gamma_i \neq \Gamma_j$, and $\underset{\sim}{\Gamma} \approx \underset{\sim}{G}$. □

Remark. It is possible to prove (although we do not do this here) a partial converse to Theorem 1. Say that R has a cofinite domain if the domain of R is a cofinite subset of N. Then if R has a cofinite domain and is finitely decomposable, L_{xy} is regular.

Theorem 2. If L is regular, the count relation R for any pair $\langle x,y\rangle$ in $\alpha x\beta y\gamma \in L$ is finitely decomposable.

Proof. Suppose L is regular. Then $|\{G_\alpha \mid \alpha \in \Sigma^*\}| < \omega$. Since $\{G_\delta \mid \delta \in \Delta\} \subseteq \{G_\alpha \mid \alpha \in \Sigma^*\}$, $|\{G_\delta \mid \delta \in \Delta\}| < \omega$. Hence by (**) and Theorem 1, $|\underset{\sim}{\Gamma}| < \omega$, and so R is f.d. □

Call the pair $\langle x,y\rangle$ ($x \neq \Lambda$, $y \neq \Lambda$) in $\alpha x\beta y\gamma \in L$ a coordinated pair iff the count relation R of $\langle x,y\rangle$ in $\alpha x\beta y\gamma$ is not f.d. Under these conditions we say that L contains a coordinated pair. So we have the following:

Corrolary 1. If L contains a coordinated pair, L is not regular.

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